

Haptic Augmentation by a Wrist Mounted Device with Laser Speckle Vibration Sensing and Hand Projection

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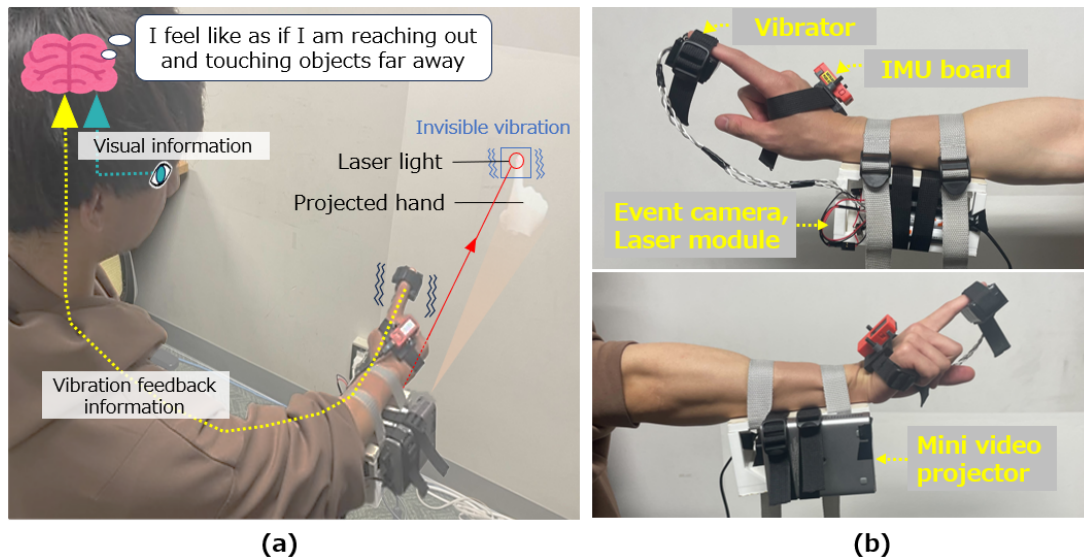


Figure 1: System Overview. (a) When a user touches an invisible vibration by projected hand with a laser, the system provides vibration feedback. The user can perceive the sensation of remotely touching the vibrated area with their finger through vibration feedback and visual feedback information. (b) Our system consists of a laser speckle vibration sensing unit (event camera, laser module), a hand projection unit (mini video projector), a projected hand control unit (IMU board), and a vibration feedback unit (feedback housing).

ABSTRACT

We propose the wrist-mounted human augmentation device with laser speckle vibration sensing and hand projection that intuitively presents the vibration characteristics of distant objects to the user. By improving the portability limitations of conventional stationary systems, our wearable device can be used while moving or performing tasks. The system consists of a laser speckle vibration sensing unit using a high-speed event camera and a projection-based extended hand via a video projector. It suppresses body sway noise through threshold processing while providing tactile feedback to the user. Additionally, the visual representation of the extended hand allows users to recognize the source of the tactile stimuli intuitively. The proposed system holds potential for various applications, such as remote object inspection.

Index Terms: Projected extended hand, Sensory addition, Remote touch sensation, Laser speckle measurement

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1 INTRODUCTION

In recent years, advancements in technologies such as artificial intelligence, robotics, and biotechnology have driven active research on “human augmentation”, which enhances human physical and cognitive abilities and enables interaction with external environments [3]. One notable area of focus is research aimed at acquiring information beyond the human senses and creating new perceptual channels. Among these efforts is research that enables users to perceive tactile information about distant objects, such as material properties, shapes, and subtle vibrations.

Previous studies have developed interfaces that allow users to perceive the material properties, shapes, and vibration characteristics of distant objects [5, 4]. For instance, Tanabe *et al.* proposed fARFEEL, a system that uses a dedicated haptic feedback device to enable users to perceive the shape of distant objects by visualizing a CG hand-arm projection (hereafter referred to as a “projected hand”) interacting with the object [5]. Similarly, Sato *et al.* proposed a method to convey the material properties of objects by adding visual effects, such as simulating finger movements on the projection-based extended hand according to the material of the object being touched, thereby inducing pseudo-haptic sensations [4].

Previous systems for perceiving the shape, material properties, and vibrations of remote objects have relied on stationary setups, limiting portability and usability. To address this, the study proposes a wearable laser speckle vibration sensing system that enables users to intuitively and accurately perceive the vibration characteristics of distant objects, even while accounting for body sway

noise. The portable design allows for use during movement or tasks without the need for a fixed setup. Furthermore, this study focuses on haptic augmentation through vibration perception, using a projection-based extended hand to convey vibration characteristics of distant objects. This approach enhances the sense of physically interacting with remote objects, creating a closer connection between the user and the environment.

The conceptual overview of the proposed system is illustrated in Fig. 1. The system uses a high-speed event camera to capture minute fluctuations in the laser speckle patterns, which occur between the laser light projected from the wearable device and the distant object. Body sway noise primarily manifests as high-intensity components in low-frequency bands or low-intensity components across all frequency bands. These noise components are removed through threshold filtering. The vibration signals, with body sway noise removed, are presented to the user via a tactile feedback device, enabling the user to accurately and intuitively perceive the vibration characteristics of distant objects.

The system projects a user-modeled extended hand onto distant objects alongside laser light, allowing intuitive association of tactile stimuli with their source. By integrating vibration perception with this visual feedback, the system enhances the tactile experience of interacting with distant objects, providing an innovative approach to vibration-based haptic augmentation.

2 PROPOSED SYSTEM

Details of the proposed system are shown in Fig. 2. Our system consists of a laser speckle vibration sensing unit (event camera, laser module, and half mirror), a hand projection unit (mini video projector), a projected hand control unit (IMU), and a vibration feedback unit (vibrator). After sensing vibrations using the laser speckle-based vibration sensing unit, the data are processed in the frequency domain. Then, a haptic feedback signal is generated and delivered to the user's actual finger.

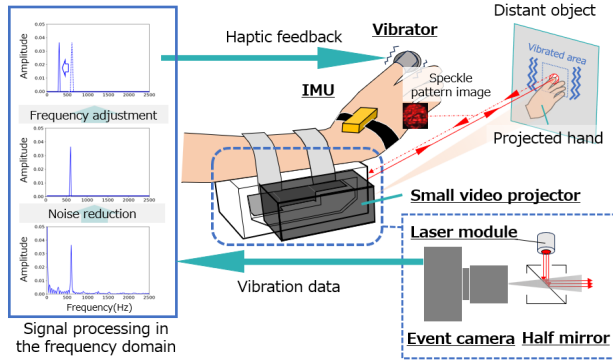


Figure 2: Details of proposed system

2.1 Laser Speckle vibration sensing

A laser is a coherent light source with aligned frequency and phase, and it generates interference fringes called laser speckles due to the fine irregularities on the surface of an object. When these interference fringes are observed with a defocused camera, the speckle pattern shifts parallelly in response to the object's vibration [6]. By leveraging this phenomenon, vibration information of a distant object can be obtained by measuring the displacement of the laser speckle image.

To rapidly capture the brightness fluctuations of each pixel caused by the parallel movement of the laser speckle image, an event camera operating with a temporal resolution of several tens of microseconds is employed. An event camera is a specialized

camera that asynchronously detects pixel-wise brightness change events, achieving high temporal resolution and low bandwidth [1]. As these characteristics, event cameras can generally measure at higher frame rates compared to standard cameras.

The event camera operates asynchronously, generating independent event sequences for each pixel. Therefore, to perform frequency analysis, it is necessary to create time series data based on a fixed sampling interval from these event sequences. Specifically, by recording the occurrence of events according to the sampling rate set within the event camera, time series data in the form of count data can be obtained. This count data can then be used for frequency analysis, where the temporal fluctuations of the data are analyzed in the frequency domain, allowing for the identification of characteristics of frequency components. Thus, the number of event data occurrences is counted every Δt seconds (sampling frequency $fs = 1/\Delta t$). Then, Fast Fourier Transform (FFT) is applied to the counted data to obtain the vibration frequency of the distant object (Fig. 3). At this time, the maximum measurable frequency is $f = fs/2$. Furthermore, since the counted data represent the magnitude of the laser speckle's fluctuation and are independent of the positive or negative direction of the fluctuation, the frequency signal derived from the count data corresponds to twice the physical vibration frequency. This allows vibration information to be obtained from changes in laser speckles without direct contact with the object.

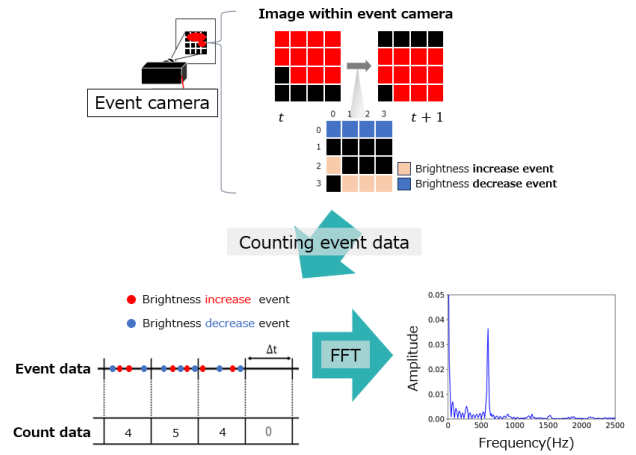


Figure 3: Vibration sensing based on speckle pattern image using event camera

2.2 Sensing method considering the inclusion of body sway noise

When the device is worn, involuntary body sway by the user can occur, resulting in event data that is not caused by the vibration of a distant object. As a result, it becomes challenging to differentiate whether the recorded event data originates from the distant object's vibration or the user's body sway (Fig. 4). This uncertainty can lead to the contamination of the measurement results with body sway noise, compromising the accuracy of the analysis. To address this, a previous study investigated the impact of the user's body sway on sensing results. The findings suggested that body sway noise appears as high-intensity signals in the low-frequency range and as low-intensity signals across all frequency ranges [2]. Therefore, thresholding is applied to remove body sway noise and construct vibration data. The processed vibration data is then intensity-adjusted and presented to the user (Fig. 5).

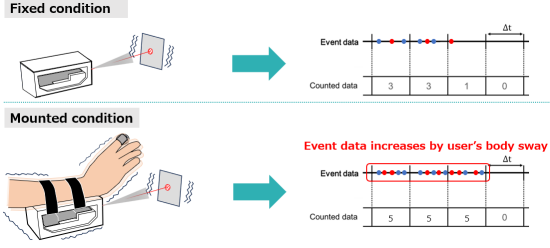


Figure 4: Fixed condition and mounted condition

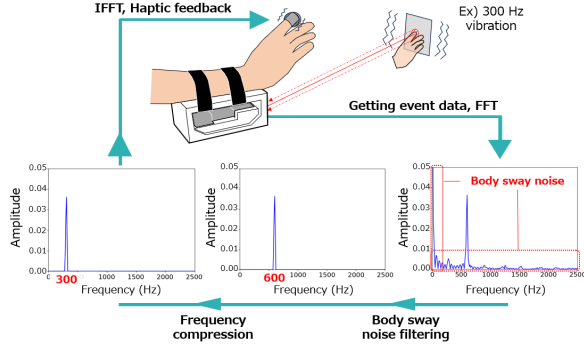


Figure 5: Vibration feedback system flow

3 SYSTEM IMPLEMENTATION

3.1 Vibration sensing unit

The vibration sensing unit consists of a red laser module, an event camera (DVXplorer by iniVation), and a half-mirror (CCM1-BS013/M by THORLABS). To align the laser beam coaxially with the event camera, a custom mount was fabricated using a 3D printer to fix the laser and camera, integrating them with the half-mirror. This configuration ensures that the positional relationship between the laser beam and the event camera remains constant regardless of the distance to the target. Furthermore, the entire sensing device is enclosed in a PLA housing, with an attachable wristband, implementing it as a wearable vibration sensing unit (Fig. 6).

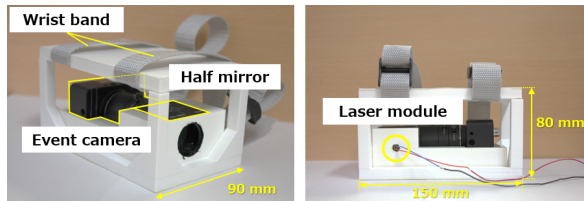


Figure 6: Implementation: Vibration sensing unit

3.2 Hand projection unit

The extended hand projection uses the mini video projector (Zenbeam E1 by ASUS), integrated with the vibration sensing unit (Fig. 7(a)). The hand model's coordinates are adjusted so that the laser light aligns with the index finger during projection.

3.3 Vibration feedback unit

The vibration feedback unit consists of a voice coil-type vibration actuator (BitTradeOne, model ADTED), a PLA housing, and a mounting band (Fig. 7(b)). When the fingertip of the projected

augmented hand (laser speckle) touches the vibrating object, vibration feedback is generated by the actuator using processed vibration data, as detailed in Section 2.2. This allows the user to perceive the vibration feedback directly through their real hand.

3.4 Projected hand control unit

To synchronize the movement of the projected augmented hand with the user's actual hand motion, an IMU sensor-equipped microcontroller (M5stickC Plus by M5stack) is attached to the user's metacarpal region (Fig. 7(c)). This setup captures the tilt of the actual hand and reflects it in the 3D hand model, which is used for projection as an augmented hand through gain adjustment and initial position calibration. In this system, pitch motion (up-and-down wrist rotation) based on the coordinate axes shown in Fig. 8 is reflected. Roll motion naturally synchronizes with the projector, requiring no corrections. Yaw motion isn't implemented due to projection distortions.

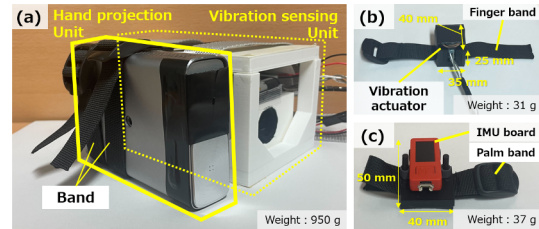
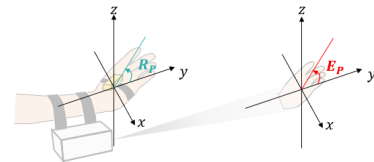


Figure 7: Implementation: (a) Vibration sensing unit with a hand projection unit, (b) Vibration feedback unit, (c) Projected hand control unit



$$E_p = R_p \times \text{gain} + \theta_p$$

R_p : Pitch angle of real hand E_p : Pitch angle of hand model before projection
 gain : adjustment gain θ_p : offset angle

Figure 8: The angular relationship between the real hand and the hand model before projection

4 SYSTEM FUNCTIONALITY VERIFICATION THROUGH EXPERIMENTS

We experimented to apply this system to an actual vibrating object and verified whether the vibrations of a distant object could be accurately fed back to the user.

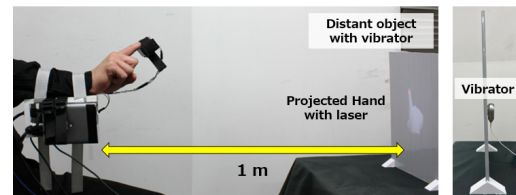


Figure 9: Experiment setup. A vibrator is installed behind the distant object.

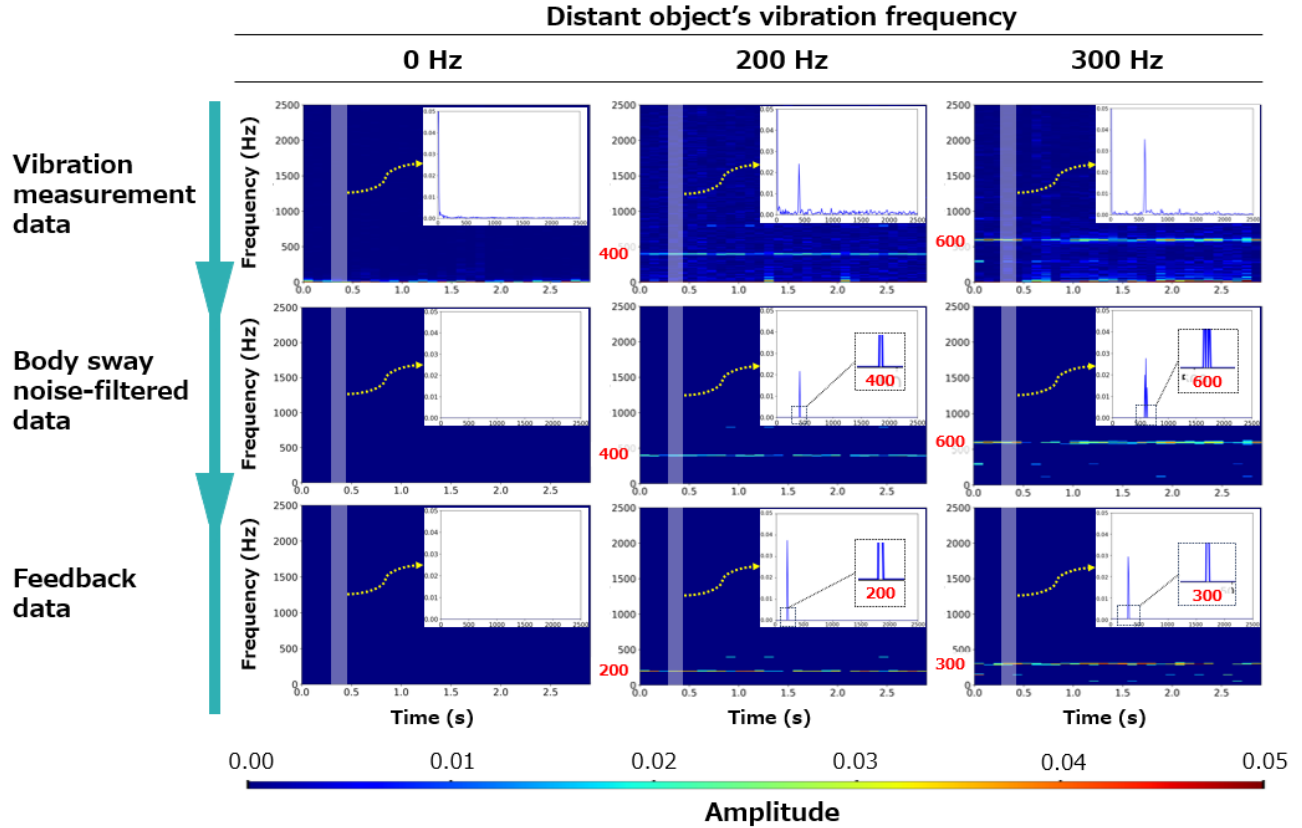


Figure 10: Color map display of the results using the spectrogram. Signal amplitude is visually represented by different colors. In each spectrogram, a 0.1-second segment of the signal spectrum is extracted and displayed to illustrate the frequency content of the signal over time. The detailed analysis reveals that, even at excitation frequencies of 200Hz and 300Hz, the system successfully generates signals that match the excitation frequencies, providing accurate and responsive feedback to the user.

4.1 Experiment setup

The experimental setup is shown in Fig. 9. A retroreflective sheet was attached to the surface of the measurement object to facilitate the observation of laser speckles. The object was vibrated using a vibrator installed behind it, and measurements were conducted from a distance of 1 meter. Sinusoidal probe signals were used to drive the vibrator at frequencies of 0 Hz, 200 Hz, and 300 Hz, each for 5 seconds. The event camera, which was used to generate count data by detecting changes in light intensity, had a sampling frequency set to 5000 Hz.

4.2 Result and Discussion

The results are shown in Fig. 10. When the object is not subjected to excitation, only the user's body motion noise appears in the measurement results. It has been confirmed that such noise can be effectively removed using the proposed method. Even when the object is excited, signals corresponding to the harmonic frequencies of the excitation signal and body sway noise can still be observed. The proposed method suppresses the body sway noise components and emphasizes the excitation frequency in the feedback provided to the user. These results suggest that the proposed method can successfully convey the vibrations of distant objects to the user while minimizing interference from body sway noise.

5 CONCLUSION AND FUTURE WORK

We proposed a wearable human augmentation system that combines laser speckle vibration sensing with a projection-based extended hand interface to intuitively convey the vibrations of distant objects. This system improves portability and flexibility by overcoming the limitations of traditional stationary setups, providing real-time haptic feedback on vibration.

Using a high-speed event camera and threshold processing, the system suppresses body sway noise while delivering accurate tactile feedback. Experiments have demonstrated that the proposed system can accurately measurement the vibration information of distant objects. The projected extended hand enables users to intuitively recognize the source of the vibration.

This device has potential applications in remote inspection with future developments aimed at expanding its capabilities to detect internal object states and micro-texture differences from vibration information, offering broad utility in both industrial and research fields.

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